PROJECT BRIEF / FULL-STACK ENGINEERING

PharmaLink

Keeping pharmacy sales and inventory in step.

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PROBLEM & USERS

Sales and inventory need to remain consistent when stock is deducted or a transaction is reversed. Staff also need visibility into product expiration and permissions appropriate to their role.

Built for: Administrators · pharmacists

MY CONTRIBUTION

I built PharmaLink independently as a school project. Its focus is the connection between pharmacy inventory, point-of-sale workflows, and role-based access.

MAIN TECHNOLOGIES

ASP.NET Core 8 / C# / React 19 / TypeScript / PostgreSQL / Dapper / JWT / Tailwind CSS

VERSION SCOPE

Documented backend and reviewed client workflows. The backend architecture is described in the supplied project documentation. Code and test references inspect client contracts and interfaces; the walkthrough simulates sale and stock behavior.

WORKFLOW

01. Review inventory

02. Record a sale

03. Reverse when needed

SYSTEM STRUCTURE

Interface: React / TypeScript

API: ASP.NET Core / JWT

Data access: Dapper

Persistence: PostgreSQL

ENGINEERING DECISION

The documented backend groups sale updates in database transactions. Reviewed client contracts keep original records, effective records, and adjustments distinct, with administrator and pharmacist workflows.

TRADE-OFF

Coordinating sales and stock in one transaction makes failure handling explicit and requires carefully defined stock and reversal rules.

READ CASE STUDY | TRY WALKTHROUGH | VIEW SOURCE

Portfolio case studies and walkthroughs provide further detail and sample interactions.